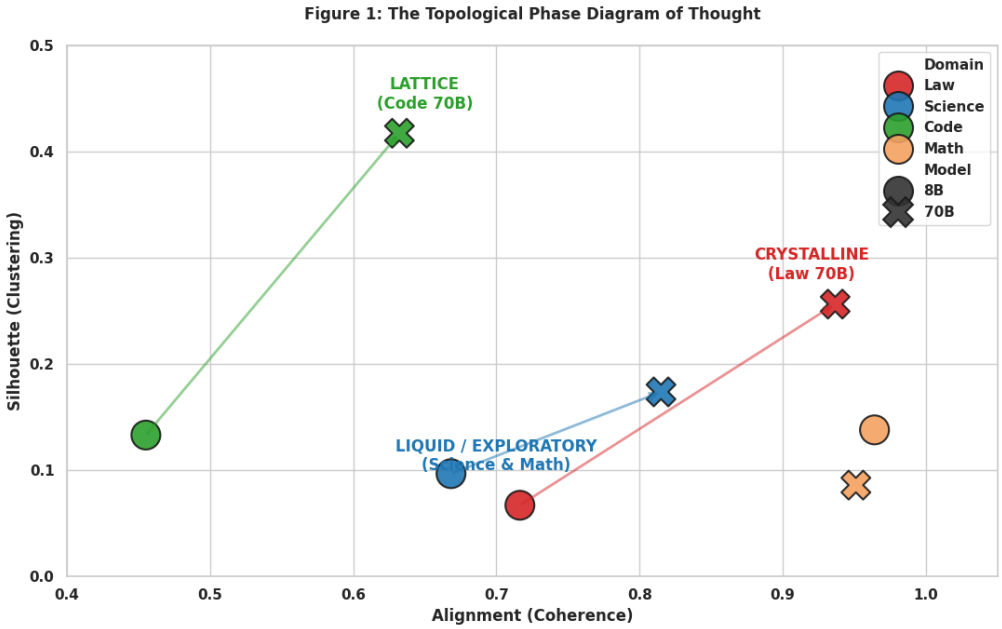


**Figure 1. The Topological Phase Diagram of Thought.** Scatter plot of Alignment vs. Silhouette Score across $N > 25,000$ reasoning trajectories. Arrows indicate the 8B→70B transition for each domain. Law (red) undergoes a phase transition from the Liquid region to the Crystalline region. Science (blue) and Math (orange) remain in the Liquid/Aligned region. Code (green) occupies a distinct Lattice region characterized by high clustering.

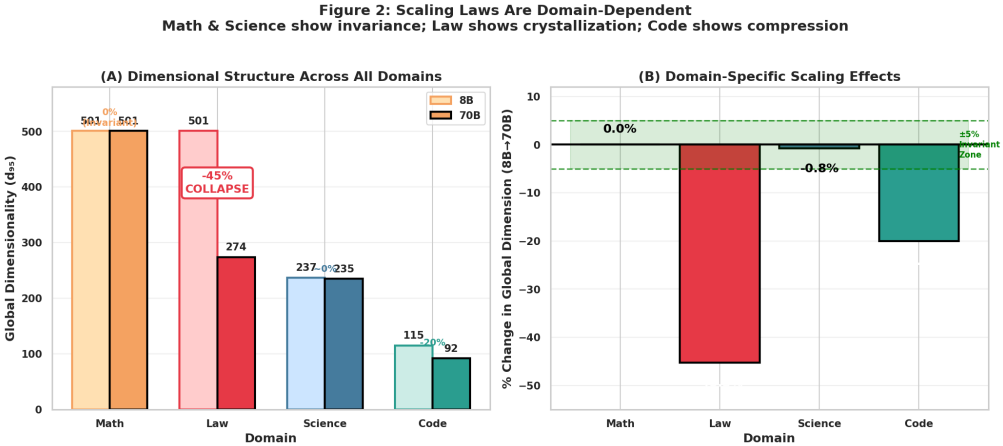


**Figure 2. Scaling Laws Are Domain-Dependent.** (A) Global dimensionality ($d_{95}$) across four domains at 8B and 70B scale. Law exhibits a 45% collapse from 501 to 274 dimensions; Math and Science remain invariant. (B) Percentage change in global dimension, with ±5% invariance zone. Law is a statistical outlier; Science and Math fall within the invariance zone.

geometry is striking: Law's slope indicates a fundamental restructuring of representational space, while Math and Science trace horizontal lines indicating that additional parameters do not reorganize how these domains are represented.

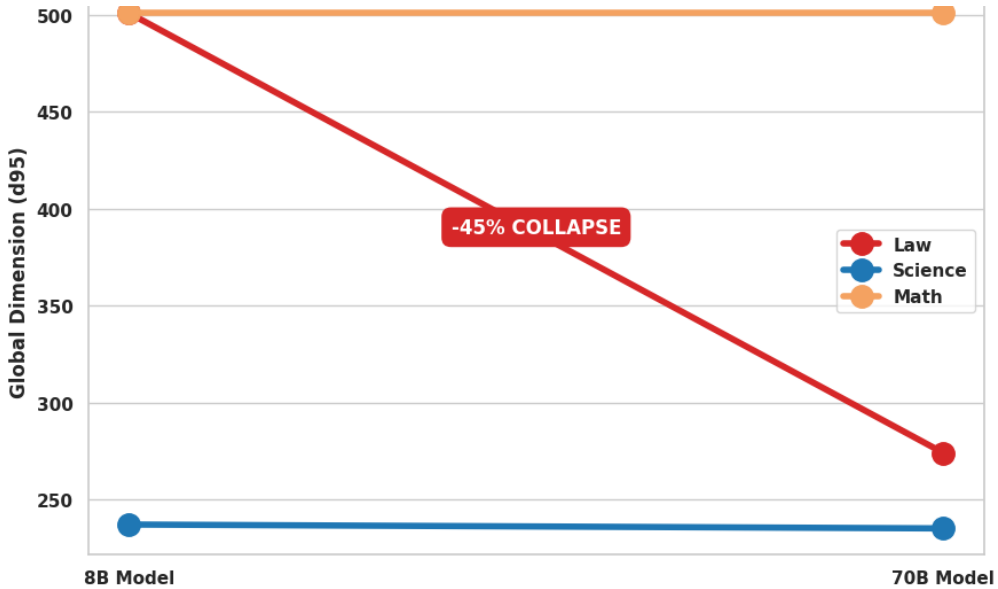


**Figure 3. The Crystallization Event vs. Scale Invariance.** Slope chart comparing global dimension ($d_{95}$) between 8B and 70B models. Law (red) exhibits a steep $-45\%$ collapse; Science (blue) and Math (orange) remain flat, demonstrating scale invariance in exploratory domains.

## Complete Geometric Characterization

Figure 4 presents the complete geometric metrics matrix: six metrics across four domains and two scales, yielding 48 measurements that comprehensively characterize the structure of reasoning.

Several patterns emerge from this visualization:

- **Alignment** (row 1) is highest for Math (0.95–0.96) and Law 70B (0.94), indicating these domains exhibit the most directionally coherent reasoning. Code shows the lowest alignment (0.46–0.63), consistent with the oscillatory, error-correcting nature of programming.
- **Silhouette score** (row 2) peaks dramatically for Code 70B (1.00 on the normalized scale, corresponding to raw silhouette of 0.42), confirming the Lattice structure. Law 70B shows elevated clustering (0.64 normalized) relative to Law 8B (0.17), consistent with Crystallization.
- **Compactness** (row 3, computed as $1 - d_{95}/550$) reveals that Code occupies the most compact manifold (0.79–0.83), followed by Science (0.57) and crystallized Law 70B (0.50). Math and Law 8B remain diffuse (0.09).
- **Intrinsic dimension** (row 4) shows remarkable consistency: all conditions fall within the 15–25 dimensional range (normalized values 0.27–0.93). Under the Levina–Bickel estimator, this indicates a stable low-dimensional structure in the sampled trajectory states across domains and scales, even as global geometry reorganizes dramatically.

- **G/L Coherence** (row 5, computed as $1/G : L$ ratio) reveals the most striking domain differentiation. Law 70B achieves perfect coherence (1.00), indicating complete manifold untangling. Code and Science 70B approach unity (0.89–0.97). Math and Law 8B remain fragmented (0.10).
- **Coherence** (row 6, step-to-step trajectory smoothness) is uniformly low across all conditions (0.07–0.15 normalized), indicating universal oscillatory dynamics—a finding we explore in Section 5.5.

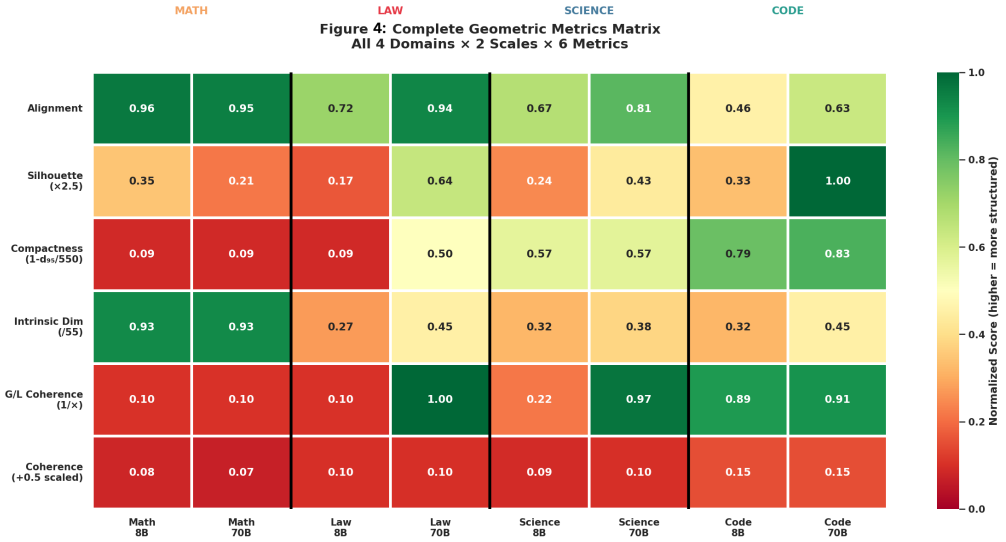


**Figure 4. Complete Geometric Metrics Matrix.** Heatmap of six geometric properties across 4 domains × 2 scales. Darker green indicates higher structure. Note the dramatic increase in structure for Law 70B compared to Law 8B across alignment, silhouette, compactness, and G/L coherence metrics.

## Mechanism of Crystallization: Manifold Untangling

What geometric mechanism drives Crystallization? We quantified manifold structure using the Global-to-Local Dimension Ratio (G/L), which measures how "tangled" or folded the reasoning trajectory is. A high G/L ratio indicates a complex, multiply-folded manifold where global variance far exceeds local neighborhood structure. A ratio approaching 1.0 indicates a flat, unrolled manifold where global and local geometry coincide.

Figure 5 presents G/L ratios across all eight conditions, revealing the mechanism of Crystallization with striking clarity.

Law exhibits the most dramatic untangling. At 8B scale, legal reasoning has a G/L ratio of 9.82×—nearly ten times more global variance than local structure, indicating a highly tangled manifold. At 70B scale, this ratio collapses to 0.98×, essentially unity. The manifold has been completely unrolled. This 10× reduction in tangling constitutes the geometric signature of Crystallization: scale enables the model to "flatten" the complex, nonlinear relationships of legal argumentation into a linear, easy-to-traverse plane.

Math presents the sharpest contrast. The G/L ratio remains locked at 9.82× at both scales—mathematical reasoning retains its complex, high-dimensional intrinsic curvature regardless of model capacity. Scale provides more parameters but does not simplify the manifold structure.